Kyle Cuss

Gameplay / Systems Engineer

Contact

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Summary

Gameplay and systems engineer with 4 years of professional experience developing games in Unreal Engine with C++ and Blueprints and Unity with C#. Experienced in owning gameplay features from initial architecture through implementation, multiplayer integration, optimization, iteration, and delivery.

Strong background in multiplayer gameplay, AI, progression, combat, UI, and data-driven systems. Comfortable working independently within existing codebases, diagnosing architectural and networking problems, and collaborating with design, art, and engineering across distributed teams. Continually expands technical knowledge through personal projects focused on exploring new tools, engines, and approaches to gameplay architecture.

Skills

Game Engines	Unreal Engine 5 (C++ / Blueprints), Unity (C#)
Gameplay Engineering	Gameplay Systems Architecture, Multiplayer Gameplay, Combat Systems, AI, Progression, Objectives, UI Systems, Data-Driven Frameworks
Unreal Engine	Replication, Dedicated Servers, GameMode / GameState / PlayerState Architecture, Behavior Trees, StateTree, AI Perception, Gameplay Ability System (GAS), Subsystems
Technical Systems	Performance Optimization, Object Pooling, Event-Driven Systems, Save and Profile Data, Backend Packet Integration, Mobile Asset Delivery
Programming Languages	C++, C#, Python, Javascript
Tools & Workflow	Perforce, Diversion, Git, Azure DevOps, Jira/Trello, Agile Development

Experience

Gameplay / Systems Engineer - RED Gaming | *2025 - Present*

Gameplay and systems engineering for Bedroom Brawl, a multiplayer Unreal Engine title built around several distinct game modes.

- Designed and implemented modular C++ and Blueprint gameplay systems supporting objectives, match flow, progression, currencies, rewards, save data, and multiple game modes.
- Built a complete single-player Tower Defense mode from scratch, including grid and placement logic, data-driven traps spawned through configurable wrapper actors, enemy waves, pathing, scoring, and supporting UI.
- Led the conversion of Tower Defense to cooperative multiplayer after the mode was well received internally, adapting its placement, traps, enemies, objectives, and match state for replicated play.
- Implemented multiplayer gameplay and replication across GameMode, GameState, PlayerState, player-owned components, and gameplay actors.
- Helped restore gameplay systems during the project’s transition from listen-server development to dedicated servers, resolving authority, initialization, ownership, and replication issues.
- Developed reusable AI systems with Behavior Trees, StateTree, and AI Perception for enemies and companion characters across multiple prototypes.
- Built persistent progression and currency systems with configurable rewards, profile data, unlocks, and save support.
- Improved runtime performance using subsystem-driven architecture, object pooling, event-driven updates, and reduced actor tick overhead.
- Collaborated with design, art, and engineering to take gameplay features from concept through implementation, testing, iteration, and integration.

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Experience

Freelance Gameplay / Systems Engineer | *2024 - Present*

Provide gameplay and systems engineering for independent Unreal Engine and Unity projects, including an NDA-protected MMO client rebuild.

- Develop and integrate gameplay, combat, UI, input, and progression systems within existing client codebases and production pipelines.
- Implement client gameplay for an Unreal Engine MMO rebuild that exchanges custom packets with an external authoritative backend.
- Build client/server gameplay flows for targeting, movement, interaction, combat, abilities, and state synchronization without treating Unreal replication as the authoritative source.
- Diagnose legacy gameplay behavior at both the code and design levels, including tracing an original melee combat bug to its cause and rebuilding it as a controlled feature because it had become integral to the original game's combat.
- Support dedicated-server and multiplayer development by resolving ownership, initialization, visibility, replication, and client-state issues.
- Deliver modular implementations designed for ongoing iteration while preserving compatibility with existing gameplay frameworks and client requirements.
- Provide architectural recommendations and technical documentation for small distributed teams.

Game Engineer (Contract) - *Wamba World* | *July 2023 - Aug 2025*

Game engineer on Waffle Smash: Diner Rush, a mobile match-3 and diner-management game released on Android and iOS. Worked on the project for approximately one year leading into launch and continued supporting it for another year after release.

- Implemented and maintained gameplay systems spanning match-3 mechanics, diner gameplay, progression, UI, tutorials, rewards, and live product requirements.
- Served as the primary engineer extending and adapting an existing third-party match-3 framework to support the game's custom mechanics and production requirements.
- Developed data-driven scene management, tutorial, and gameplay systems using Scriptable Objects and dependency injection.
- Integrated Unity Addressables with Google Play Asset Delivery to support on-demand asset streaming.
- Reduced memory usage and initial download size through dynamic asset loading, unloading, and content packaging improvements.
- Maintained the Android build pipeline and supported Google Play Store publishing through launch and post-launch updates.
- Led Unity engine upgrades and resolved package, plugin, and dependency conflicts across the project.
- Worked closely with design and art to prototype, evaluate, and refine features throughout production and post-launch support.

Gameplay Programmer (Voluntary) - *Greatsword Games* | *2023*

Contributed to an Unreal Engine MMORPG prototype.

- Refactored prototype Blueprint ability systems into C++ using Unreal Engine's Gameplay Ability System.
- Improved multiplayer replication reliability and prepared gameplay systems for continued expansion.
- Collaborated with the engineering team to move prototype features toward a more maintainable production architecture.

Projects

Technical project breakdowns and case studies available at:
https://cussk.github.io/kylecuss.com

Education

Unity *Dec 2022*	**Unity Learn** *Unity Junior Programmer, Unity Creative Core, Unity Essentials*
Skills for Hire Atlantic *Dec 2022*	**Web Development Bootcamp** *Certificate of Completion*
McMaster University *2011*	**Batchelor of Arts** *Anthropology*